Qatar-1b

R-0059

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_201031` · created 2026-07-15T14:08:28+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459153.67 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.014
Transit depth (Rp/Rs)^2	3.7 +/- 0.53 [%]
Orbital Inclination (inc)	84.09 +/- 1.95
Airmass coefficient 1 (a1)	0.953 +/- 0.023
Airmass coefficient 2 (a2)	0.03 +/- 0.02
Scatter in the residuals of the lightcurve fit is	2.41 %
Best Comparison Star	#1 - [461, 179]
Optimal Method	PSF photometry
Transit Duration (day)	0.07 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.014 vs 0.1463 (deviation 2.73σ)
Duration fit vs published	0.07 d vs 0.0692 d (deviation 0.05σ)
Mid-time O–C	-4.9 min
Depth SNR	11.5
Residual scatter	2.41 %

Light curve

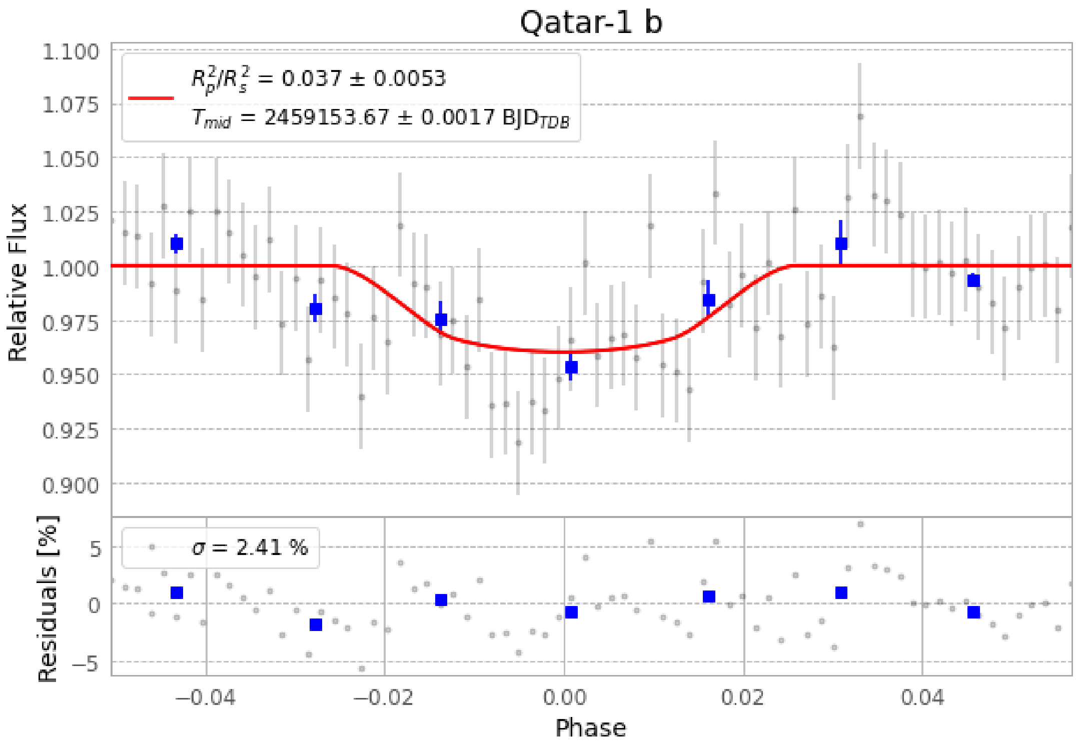

Figure 1 — FinalLightCurve_Qatar-1 b_2020-10-30.png (SHA-256 029937369026461c…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [466, 372], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 ac485b246b88b09d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

74 FITS frames (49 MB), `Qatar-1201031021812.FITS` → `Qatar-1201031055712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-201031.log (268ff1ba25f1…), FinalLightCurve_Qatar-1 b_2020-10-30.png (029937369026…), FinalLightCurve_Qatar-1 b_2020-10-30.csv (35655c98cb9d…)

Compute resources

Wall time 172.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-15 16:40Z.